Neel Shah

Data Engineering · Analytics · Business Intelligence

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github.com/neeljshah/court-vision | linkedin.com/in/neeljshah22

SUMMARY

I build data pipelines, reporting and the checks that keep the numbers honest. At Sunsolor I built live **Tableau / Power BI** suites and automated daily market-trend pipelines in **Python**; at Fortrex Securities I built executive reporting interfaces and **reporting pipelines**. In my project, CourtVision, I built a **PostgreSQL** schema with a **SQLite** fallback, a **151-file intelligence layer** including a **291,625-pair** matchup matrix, and a validation harness that caught and retracted six of my own inflated results.

PROFESSIONAL EXPERIENCE

Data Analyst Intern, Sunsolor · *Tempe, AZ · 2025*
Built live **Tableau / Power BI** suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in **Python**; **data-quality audits** across 5+ datasets; optimized **SQL** on **1M+-row** datasets.

Data Annotation Specialist, DataAnnotation.Tech · *Remote · 2024–2025*
Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored corner-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst, Fortrex Securities · *Madison, WI · 2023–2024*
Built secure executive reporting interfaces (**99.9% uptime**); refined backend payment APIs & **reporting pipelines** for high-throughput billing; engineered real-time **anomaly-detection** alerting.

SELECTED PROJECT: COURTVISION

Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.
Evidence: github.com/neeljshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neeljshah/court-vision/blob/master/EVIDENCE.md

Data engineering

- **Persistence: PostgreSQL** schema (12 core tables) with a transparent **SQLite** fallback that auto-translates PG **SQL**, plus tracked, idempotent migrations.
- **Evidence site:** one unified, CI-gated site at https://neeljshah.github.io/court-vision/, built by one person directing AI coding agents; performance-budget and accessibility (axe) gates are hard CI steps with zero serious or critical violations; a PAPER ONLY desk record with receipts.
- **Testing and delivery:** ~7,400-test **pytest** suite across ~580 files, CI/CD on 3 GitHub Actions workflows, and 5 Docker images; a 430-module codebase with research code kept apart from the load-bearing deployment graph.
- **API and services:** FastAPI serving layer of ~99 endpoints across 12 routers (REST, WebSocket, SSE) and server-rendered Jinja dashboards; 9 long-running daemons under a watchdog supervisor.

Analytics and validation

- **Pregame vs the devigged close:** NBA moneyline Brier 0.1735 vs 0.1666 for the close (n=372 held out, 95% CI of the gap includes 0); MLB moneyline 0.2429 vs 0.2390 (n=13,992); soccer O/U 2.5 0.2465 vs 0.2390 (n=7,558); totals and ATP tennis trail by more. Nothing beats the close.
- **Player props (NBA):** chronological holdout of 20,354 player-games, MAE PTS 4.83, REB 1.92, AST 1.39, FG3M 0.89; reported as accuracy only, with no market comparison.
- **Harness:** expanding-window **walk-forward** with a per-fold date assertion, truncation-invariance leak tests, acceptance only when a change beats baseline on two independent corpora, and a published minimum detectable effect for every market-relative row.
- **Published results:** pregame probabilities match or trail the devigged close and the in-game model is behind the live market, with n and 95% intervals; six withdrawn figures published with reasons at /evidence/retractions/; 16 findings on one template with limitations and verdict history.

EDUCATION & CERTIFICATIONS

University of Iowa: three years of data-science coursework. Coursework: math stats, probability theory, machine learning, financial analytics.
Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

Languages: Python (expert), SQL, JavaScript/TypeScript (working)
Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau, Power BI, dashboards
ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels
Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration